

Literature Status Packet: $L_2 + L_\infty$ versus $L_2 \cap L_\infty$

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Status

The final problem in Astashkin–Maligranda arXiv:1704.01717 asks whether

$$L_2(0, \infty) + L_\infty(0, \infty) \quad \text{and} \quad L_2(0, \infty) \cap L_\infty(0, \infty)$$

are isomorphic. This problem has a later negative answer: Astashkin–Maligranda arXiv:1804.03469 proves that these two spaces are not isomorphic.

This packet is classified as `literature.already_answered`, not as an original solution.

Original Problem Source

Original source:

- S. V. Astashkin and L. Maligranda, $L_p + L_\infty$ and $L_p \cap L_\infty$ are not isomorphic for all $1 \leq p < \infty$, $p \neq 2$, arXiv:1704.01717; Proc. Amer. Math. Soc. 146 (2018), no. 5, 2181–2194.

The question appears on page 13:

4 Concluding remarks related to the spaces $L_2 + L_\infty$ and $L_2 \cap L_\infty$

We do not know whether the spaces $L_2 + L_\infty$ and $L_2 \cap L_\infty$ are isomorphic or not.

Problem 2. *Are the spaces $L_2 + L_\infty$ and $L_2 \cap L_\infty$ isomorphic?*

We end up the paper with the following remarks related to the above problem.

Remark 2. *The predual spaces $L_1 \cap L_2$ and $L_1 + L_2$ for $L_2 + L_\infty$ and $L_2 \cap L_\infty$, respectively, are not isomorphic.*

Supporting Answer Source

Supporting source:

- S. V. Astashkin and L. Maligranda, $L_p + L_q$ and $L_p \cap L_q$ are not isomorphic for all $1 \leq p, q \leq \infty$, $p \neq q$, arXiv:1804.03469.

The abstract explicitly identifies the $L_2 + L_\infty$ case as an answer to a question in reference [2]:

**$L_p + L_q$ and $L_p \cap L_q$ are not isomorphic
for all $1 \leq p, q \leq \infty, p \neq q$**
 **$L_p + L_q$ et $L_p \cap L_q$ ne sont pas isomorphes pour tout
 $1 \leq p, q \leq \infty, p \neq q$**

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Abstract

We prove that if $1 \leq p, q \leq \infty$, then the spaces $L_p + L_q$ and $L_p \cap L_q$ are isomorphic if and only if $p = q$. In particular, $L_2 + L_\infty$ and $L_2 \cap L_\infty$ are not isomorphic which is an answer to a question formulated in [2].

Résumé

Nous prouvons que si $1 \leq p, q \leq \infty$, alors les espaces $L_p + L_q$ et $L_p \cap L_q$ sont isomorphes si et seulement si $p = q$. En particulier, $L_2 + L_\infty$ et $L_2 \cap L_\infty$ ne sont pas isomorphes, ce qui est une réponse à une question formulée dans [2].

The supporting paper's Theorem 2 states the exact non-isomorphism:

$L_2 + L_\infty$ and $L_2 \cap L_\infty$ are not isomorphic

Let x be a measurable function on $(0, \infty)$ such that $m(\text{supp } x) \leq 1$. Then, clearly, x is equimeasurable with the function $x^* \chi_{[0,1]}$. Therefore, assuming that $x \in L_2$ (resp. $x \in L_\infty$), we have $x \in L_2 + L_\infty$ and $\|x\|_{L_2+L_\infty} = \|x\|_{L_2}$ (resp. $x \in L_2 \cap L_\infty$ and $\|x\|_{L_2 \cap L_\infty} = \|x\|_{L_\infty}$).

Theorem 2. *The spaces $L_2 + L_\infty$ and $L_2 \cap L_\infty$ are not isomorphic.*

Proof. On the contrary, assume that T is an isomorphism of $L_2 + L_\infty$ onto $L_2 \cap L_\infty$.

For every $n, k \in \mathbb{N}$ and $i = 1, 2, \dots, 2^k$ we set

Its reference [2] is the original 2018 PAMS paper:

- [1] J. Bergh and J. Löfström, *Limits in Banach Space Theory*, Springer-Verlag, New York 2000.
- [2] S. V. Astashkin and L. Maligranda, $L_p + L_\infty$ and $L_p \cap L_\infty$ are not isomorphic for all $1 \leq p < \infty, p \neq 2$, Proc. Amer. Math. Soc. 146 (2018), no. 5, 2181–2194.
- [3] C. Bennett and R. Sharpley, *Interpolation of Operators*, Academic Press, Boston 1988.
- [4] J. Bergh and J. Löfström, *Interpolation Spaces. An Introduction*, Springer-Verlag, Berlin-New York 1976.

Verification Notes

- The original paper states the $L_2 + L_\infty$ versus $L_2 \cap L_\infty$ question as open in Section 4.
- The supporting paper is separate and later. It says that the $p = 2, q = \infty$ case was the motivation to continue the earlier work.
- The supporting theorem is stronger than needed: its main theorem gives non-isomorphism of $L_p + L_q$ and $L_p \cap L_q$ for all distinct $p, q \in [1, \infty]$.

Search Bounds and Limitations

Before promotion, the run registry, solution index, attempt index, proof-gap index, and local parsed source for arXiv:1704.01717 were checked for duplicate coverage. Web searches on 2026-06-13 for the original title, arXiv id, and the $L_2 + L_\infty / L_2 \cap L_\infty$ phrases located arXiv:1804.03469 as the direct later answer.

This packet does not address the separate Problem 1 in arXiv:1704.01717 asking whether $L_1 + L_\infty$ is isomorphic to a dual Banach space.

Human Review Recommendation

Check the four displayed evidence images: the original question, the supporting abstract’s answer statement, the supporting Theorem 2, and the reference-list identification of [2]. Together they establish the exact literature-status transfer.